# ENGINEERING GRAPHICS

**I Semester (AERO, EEE & MECH)**

| Course Code | Category | Hours / Week | | | Credits | Maximum Marks | | |
|---|---|---|---|---|---|---|---|---|
| | | L | T | P | C | CIE | SEE | Total |
| A6ME01 | ESC | 1 | - | 4 | 3 | 40 | 60 | 100 |

## Course objectives:

The course should enable the students to:
1. Create awareness and emphasize the need for Engineering Drawing in various branches of engineering.
2. Enable the student with various concepts of dimensioning, conventions and standards related to engineering drawings.
3. Follow the basic drawing standards and conventions.
4. Develop skills in three-dimensional visualization of engineering component.

## UNIT-I INTRODUCTION TO ENGINEERING GRAPHICS

Introduction to Engineering Graphics: Principles and their significance,
Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, Modify Commands, 2D Drawings-Simple Exercises.

## UNIT-II ENGINEERING CURVES

Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only).
Special curves: Cycloids, Involutes.

## UNIT-III ORTHOGRAPHIC PROJECTIONS OF POINTS, LINES AND PLANES

Principles of Orthographic Projections – conventions – First and Third angle projections. Projections of points, Projection of lines, Lines inclined to both the planes.
Projections of Planes: Projections of regular planes inclined to both planes.

## UNIT-IV PROJECTIONS OF SOLIDS AND DEVELOPMENT OF SURFACES

Projection of Solids**:** Regular Solids inclined to both planes
Development of Surfaces: Theory of development, development of lateral surface with base.

## UNIT-V ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS - CONVERSION

Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections. Drawing Isometric circles, Dimensioning Isometric objects.
Conversion of Isometric view to Orthographic views and Orthographic views to isometric view.

## Text Books:


1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.


## Reference Books:


1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications
2. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C. , Engineering Drawing and Computer Graphics, Pearson Education.

**Course Outcomes : Students should be able to:**

At the end of the course the student should be able to:

1. Explain various commands and create drawing in AutoCAD.
2. Sketch the various curves used in engineering applications
3. Prepare orthographic projections of lines, planes by visualizing them in different positions.
4. Solve the problem of projections of solids and development of surfaces for industrial needs.
5. Construct the isometric view into orthographic views and vice versa